HP428 – Randomised evaluations of health programmes

2022/23 Academic Year

CANDIDATE NUMBER: (Five digit number available via LSE for You. This is not the same as your LSE Student ID)	5 1 6 0 0
ASSESSMENT TITLE: (E.g. Title of essay(s) attempted)	**Innovative group antenatal care (ANC): Protocol for a cluster-randomised control trial in Queensland, Australia**
WORD COUNT: Excludes references and appendices.	**4277**

Candidate Declaration:
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Innovative group antenatal care (ANC): Protocol for a cluster-randomised control trial in Queensland, Australia

1. Background

1.1. Adverse birth outcomes as a public issue

Adverse birth outcomes, including preterm birth, low birth weight, stillbirths and miscarriage, have a far-reaching devastating impact on individuals, families, and communities (1–3). The World Health Organisation called for global action on preterm death and other adverse birth outcomes (4). Although being one of the safest birthplaces, Australia had higher rates of preterm birth (8.3%) and low birth weight (6.5%) in 2020 compared to the OECD average of 8.0% and 5.9%, respectively (5,6). Since the early 1990s, there has been a shift in Australia from private to public obstetric care. The poor birth outcomes drive the need for a special care nursery (SCN) or neonatal intensive care unit (NICU), increasing medical costs (7,8). This exacerbates the burden on the public healthcare system with a decreasing number of general practitioners (9,10). Thus, adverse birth outcomes are not only a family tragedy but also a societal issue that requires effective intervention.

Underlying the adverse birth outcomes are often complicated social disparities. Pregnant females face challenges in both physical and mental health and require social support, which is associated with birth outcomes (2). Aboriginal, refugee and immigrant women face a higher risk of poor birth outcomes, which is associated with poor health literacy and difficulty in continuous access to healthcare (11). Suboptimal care is associated with 10-60% of stillbirths and neonatal deaths in high-income countries (12). In Australia, the models and quality of care vary according to the hospital (13,14). Despite a universally funded public health system, females with socioeconomic disadvantages are more likely to attend public clinics and have poorer satisfaction (15). Thus, a cost-effective way to improve care quality in the public health system is needed to narrow the gap in the quality and satisfaction of ANC.

1.2. Group vs. traditional antenatal care (ANC): current knowledge

High-quality ANC is a proven way to improve birth outcomes and a fundamental right of all women (16). Providing high-quality ANC is a global aim set by the United Nations and the Australian Government (17). For mothers in Australia, there is a diversity of choices for ANC, provided by either public or private healthcare providers (17,18). Most ANC models in Australia are traditional ANC, which is deeply rooted in offering one-on-one care to women to build a rapport of trust and confidence between the midwife and the woman (19). This is achieved by ensuring that the same midwife provides care throughout the entire process of pregnancy, labour, and early postnatal period (19).

In contrast, group ANC consists of group discussion among 8-12 pregnant women with similar gestational weeks plus an obstetric provider and a host, with prenatal check-ups and health education provided in group settings (20,21). It is expected to bring about more positive experiences for pregnant women, with women benefiting from both the expertise of their HCP and the knowledge, experience, and support of their peers, thus improving continuity of care (22) and birth outcomes (23). First proposed in the United States in the 1990s, CenteringPregnancy has been the most popular group ANC model, which is shown to prevent adverse birth outcomes in randomised controlled trials (RCTs) (24). Recent RCTs on other group ANC models, including Moms2B (25) and Pregnancy Circles (26), also show promising results in reducing adverse birth outcomes and improving care satisfaction.

However, there is so far no evidence from direct comparison between ANC models in Australia (13,14). Although the implementation of CenteringPregnancy in Australia indicates high acceptance among pregnant women and midwives, these studies do not provide evidence for health improvements (19,27,28). A recent meta-analysis for group ANC only shows significant improvements in mothers' depressive symptoms but no or only limited improvement in birth outcomes (21,29). Thus, there is a need to acquire clinical evidence and translate it to optimise care delivery (17). Also, the cost of group ANC varies according to the setting (30), which needs to be determined in Australia.

1.3. Proposed intervention: Expect with Me

Expect with Me is an innovative model of group ANC that combines a specialised web platform and best practices for group ANC. With the support from a reproducible intervention plan and a web platform designed to enhance patient involvement and assistance, promote healthy habits and informed choices, establish connections between patients and healthcare professionals, and improve healthcare provision, the model is designed to be scalable nationally at a reduced cost to local healthcare providers (31). Thus, it is a promising model to improve adverse birth outcomes, which better meets Australia's national standard for high-quality ANC, i.e. female-centric, safe, and supportive (32).

Furthermore, although implementing the care model may require extra investment in staff training, room hiring and intervention delivery, the prevented adverse birth outcomes may reduce the need for healthcare resources, especially SCN/NICU admissions and patient education, thus reducing the overall health cost (24). Expect with Me includes a rigorous multi-method evaluation plan to examine the impact of Expect with Me on maternal health and birth outcomes and identify and address barriers to national scalability (31), which allows data collection and reporting at a relatively low cost and in a transparent way.

1.4. Objectives

This study will compare the effectiveness of group ANC using the Expect with Me programme with traditional ANC models in terms of clinical outcomes, participant satisfaction, and cost. The results will provide important insights into the cost and effectiveness of this innovative model of group prenatal care and inform policy decisions for improving ANC in Australia.

2. Intervention and Conceptual Framework

The intervention details of Expect with Me have been described by Cunningham et al. (2017)(31). The programme provides structured training through group sessions for 8-12 women each time, consistent with most group ANC models (20,33). The content of each session is summarised in Appendix 1. The intervention content is based on best evidence and practice of existing group ANC models and integrated with a web platform for optimal participant engagement. The control group will receive standard care in the public healthcare system.

2.1. Local context

The study will be conducted in Queensland, a state that accounts for roughly 20% of births in Australia (36) and experiences inferior birth outcomes compared to the national average. In Queensland, 20.2% of newborns are admitted to SCN/NICU, exceeding the national average of 18.2%. Additionally, the state's preterm birth rate is 9.0% and low birthweight rate is 6.8%, both surpassing the national averages of 8.3% and 6.5%, respectively (5). These statistics highlight the need to enhance birth outcomes and reduce SCN/NICU admissions in Queensland.

Over 70% of mothers in Queensland receive ANC through the public healthcare system, while around 20% obtain care from private healthcare providers (37). As part of the Australian public healthcare system, Queensland Health serves as the sole provider of public health services and regulator of local private healthcare services (5). The state offers convenient access to high-quality care during confinement, even for disadvantaged groups. This underlines the importance of improving primary care (38), especially ANC, and minimises variation during confinement, allowing for more unbiased insights into the effectiveness of ANC. However, similar to other Australian states and territories, Queensland is faced with shortage in funding and staff for the public health system (39–41). Thus, the health system needs a cost-effective ANC to maintain its high standards in birth outcomes.

The feasibility and acceptability of the proposed trial in Queensland are supported by prior qualitative research demonstrating wide acceptance of group ANC among expectant women and midwives (28). Successful examples of state-wide interventions in Australia (34) indicate substantial potential for effective implementation of such programmes in Queensland. The high internet penetration rate among women of childbearing age (> 89.9% (35)) also enables them to use the online components of Expect with Me.

Queensland's diverse population, encompassing various ethnicities, socioeconomic backgrounds, and geographic regions (42,43), ensures that the trial within the state's health system will provide valuable insights into the intervention's generalisability across other Australian states and territories. National organisations dedicated to improving birth outcomes, such as the Australian Preterm Birth Prevention Alliance (44), will ensure that clinical evidence gained from this trial are disseminated throughout Australia. This collaboration will contribute to a wider impact within the national health system.

2.2. Theory of change

<u>Beavioural assumptions</u>

The theory of change is constructed by consulting a series of qualitative research on women’s experience in group ANC and quantitative research on clinical outcomes (28,31,45–47)(Figure 1). We aim to introduce behavioural change by introducing group ANC, which is a transformation of care provision. Expect with Me is a group ANC model with detailed session plans and clear take-home messages, which prolongs ANC duration.

We assume that, compared with traditional one-on-one consultations, group sessions provide a more supportive environment to address other basic, safety, and social needs during ANC, by facilitating community building, positive environment for learning and problem-solving, decreased hierarchy between healthcare providers and women. This increases the duration and continuity of ANC and enables self-assessment and self-management, contributing to optimised care experience.

The optimised care experience contributes to improved knowledge and satisfaction among women, which are associated with improved birth outcomes. Pregnant women will benefit from the knowledge of healthy behaviours rather than fear and avoidance for themselves and their families. These behavioural changes create a supportive environment for peer learning and information sharing, which empowers women and improve their satisfaction. The transformed care provision improves care knowledge in the supportive environment. Better care experience and knowledge both contribute to improved adverse outcomes and reduces healthcare costs.

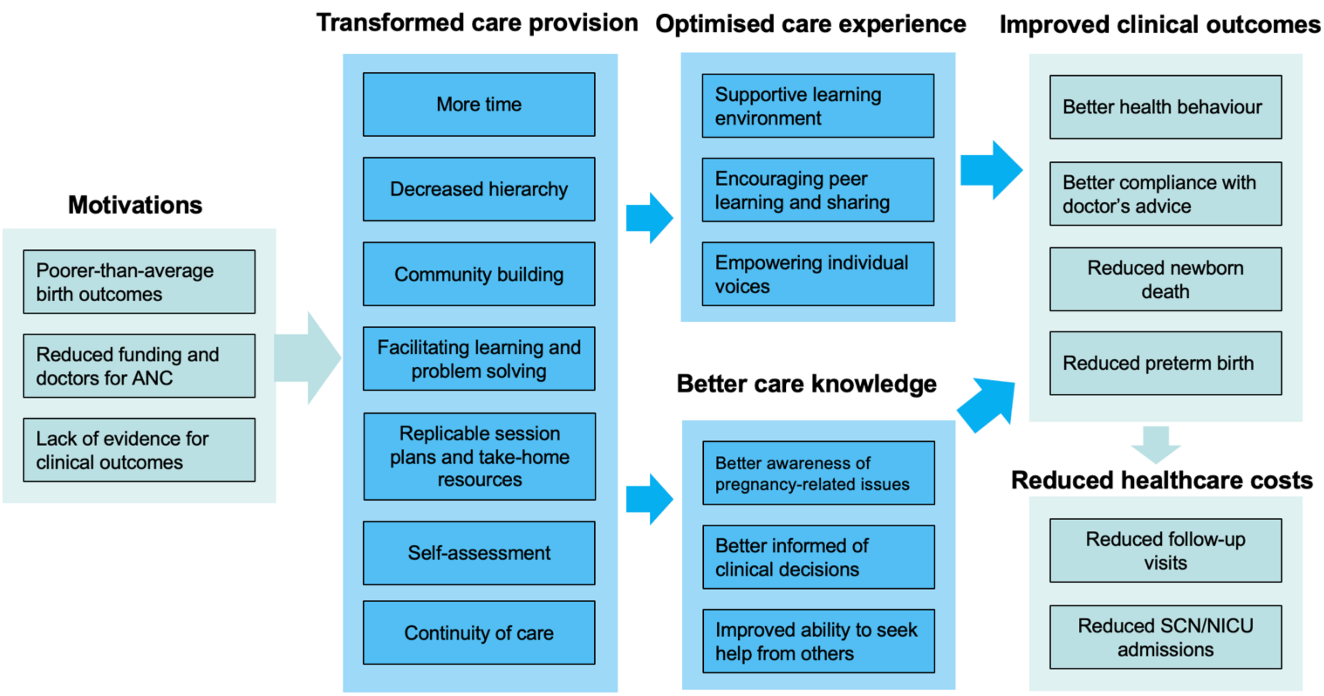

Figure 1. Theory of Change for Group ANC

<u>Implementation assumptions</u>

To implement this intervention, we assume that the ANC providers in the public health system agree to participate in the trial. Secondly, we assume that there are sufficient pregnant women of similar age in each community. We also assume that they attend all the sessions. The two assumptions ensure the sample size in the trial. Third, we assume that recruited women in each community receive the same ANC throughout the trial and do not attend clinics in other communities. Thus, we ensure the fidelity to the assigned intervention. Fourth, we assume that the participants are aware of the trial and are willing to achieve learning outcomes and peer support. Thus, we ensure the trial is implemented as per our theory of change.

3. Experimental Design

3.1. Procedures

Cluster randomisation will be performed at the community level, with a lottery design to ensure fairness. As the trial involves significant educational components, participants may spread their knowledge across intervention groups; thus, cluster randomisation is used to minimise the contamination bias (48). Randomisation will be performed by a computer programme, supervised by a researcher who does not participate in the intervention to ensure fairness (49).

The programme will be delivered at a community health facility (e.g. community clinics, community centres, general practices, and any other sites providing publicly-funded ANC). Participating communities will be randomly assigned into an intervention arm, which receives group ANC, and a control arm, which receives usual care. The parallel-group design is suitable for measuring long-term impacts and irreversible outcomes (49).

Previous data show that SCN/NICU admissions are affected by the proportion of indigenous mothers, smoking status, and caesarean sections (5). Thus, stratification will be done according to the factors, which will increase the power of the study and enable an analysis of how the intervention affects different subgroups. To minimise poststratification patient selection bias (50), we request recruited subjects to agree to participate in group ANC if assigned to their community, yet we do not confirm the assignment of group ANC, until randomisation is done.

3.2. Sample size calculation

A longitudinal cohort study using Expect with Me in the United States shows that the programme reduced NICU admission by 36% (9.4% vs. 14.6%; RR = 0.64, 95% CI: [0.49, 0.78]) (51), although the study did not report SCN admissions data. Thus, the reduction should be higher than 36%. A retrospective cohort study in the United States suggests that group ANC using CenteringPregnancy reduced SCN/NICU admission by 47% (10.2% vs. 21.3%; RR = 0.53, 95% CI: [0.36, 0.79]) (52). Thus, we conservatively assume a 35% reduction, reducing the SCN/NICU admissions rate from 20.2% to 13.1%.

We consulted relevant literature to estimate the intracluster correlation coefficient (ICC) and dropout rate in Queensland. Wiggins et al. (2020) assumed an ICC of 0.05 in the English health system (33), while Ickovics et al. (2016)(53) used an ICC of 0.001 in New York community service settings (54). Two trials in African countries both assumed ICC to be 0.01, yet one found the actual ICC to be only 0.00063 (54,55). Thus, we conservatively assume ICC to be 0.05, given the similarities between English and Australian health systems. Based on RCTs in high-income settings (31,33,53), we assume a dropout rate of 10%.

We contacted the Queensland Health for the availability of the facilities. Within Queensland Health, there are currently 108 clinical sites providing free ANC, providing care to an average of 431 mothers per year. Further sites can be sought through room hire at local neighbourhood centres (56) and general practices, which may provide more flexibility for the intervention to be delivered at the community level (see Appendix 3).

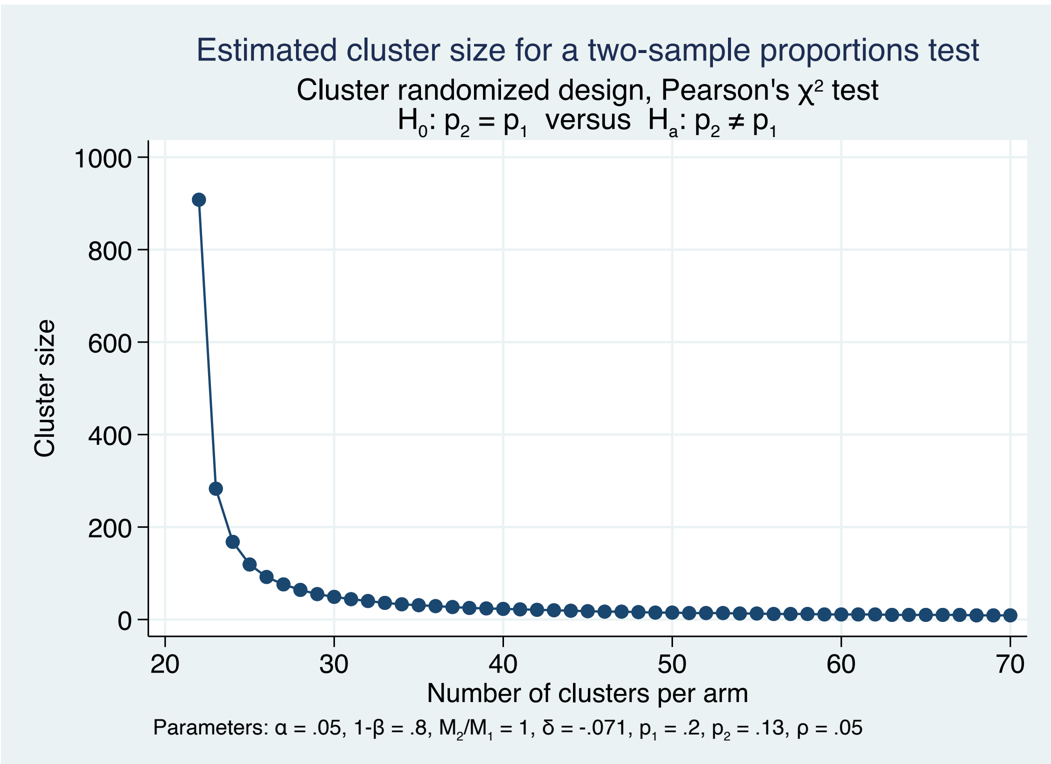

Figure 2. Estimated cluster size, given different cluster number assumptions

Assuming the significance level of 0.05, and the power of 0.8, we calculate and visualise the number of participants needed in each cluster in Stata, given different assumptions of cluster numbers (Figure 2). Assuming there be 40 sites per arm, there will be 23 participants required at each site. Considering dropouts, each site will need to recruit 26 mothers, with a total of 2,080 mothers. This means there will be three groups of 8-9 mothers recruited at each site, which is reasonable for group ANC.

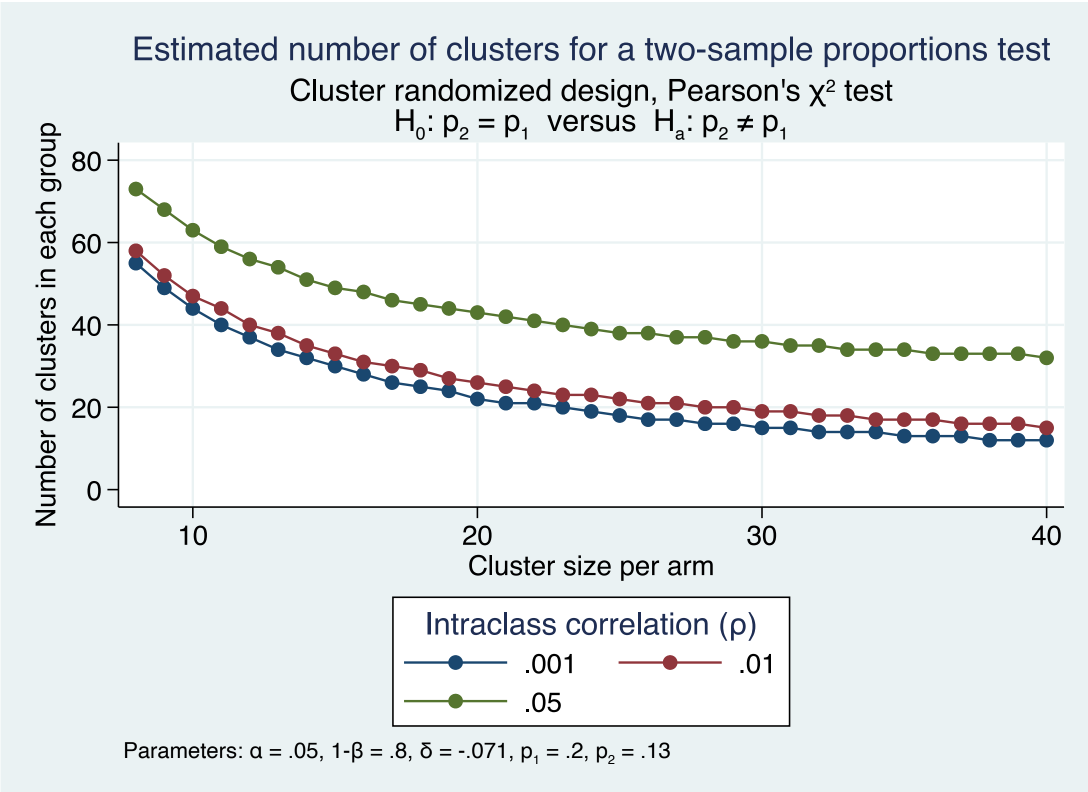

Figure 3. Estimated cluster number, given different cluster size assumptions

However, we may overestimate the sample size due to our conservative estimation of the effect size and ICC (Figure 3). To enable more accurate sample size calculation, we will perform a pilot study, to identify the number of available facilities and eligible participants, to measure the sociodemographic characteristics of the target population and to calculate the ICC. Thus, the required sample size is likely to be reduced.

3.3. Challenges to Randomisation

Partial compliance is expected to happen. Interventionists will first receive training at the University of Queensland, which had experience in implementing group ANC (28), to ensure the homogeneity and fidelity of the intervention. The proposed pilot study will be a chance to further determine the content and mode of interventionist training to ensure interventionist compliance. For participants, partial compliance is often caused by missed sessions due to illnesses, work, childcare responsibilities, etc., of which the impact may be alleviated through uploading materials on the web platform of Expect with Me.

Attrition may compromise randomisation. It is often caused by abortion, moving out of the region, and escalation of health issues (26). Thus, we consider a 10% dropout rate in recruitment to ensure sufficient power. Similar studies show that sometimes hospital midwives may encourage the participants to switch back to standard care (26). Thus, the informants about the trial should also include non-participating midwives at the participating sites.

Contamination bias may still exist because certain participants may communicate with each other online or face-to-face. However, cluster randomisation is used to minimise the effect. To detect potential spill-over, the participants will be required to answer whether they have discussed the intervention with other people in each participant survey during the trial.

4. Data Collection

4.1. Outcomes of interest

Primary outcome

The primary outcome of this trial will be the SCN/NICU admission rate, which is identified by the stakeholders to be highly relevant to both clinical outcomes and the cost-effectiveness of the trial. SCN/NICU admissions are the most common consequence of adverse birth outcomes. In Australia, 79% of preterm birth cases and 79% of low birthweight cases are admitted into SCN/NICU. Meanwhile, healthcare cost at SCN/NICU constitutes a major proportion of the direct medical cost arising from adverse birth outcomes (57,58).

Secondary outcomes

The following outcomes are considered relevant:

- Detailed birth outcomes, including gestational age, birth weight, death, are used to assess safety of the intervention in more perspectives;
- Postpartum healthcare, including frequency and duration of follow-up visits, are used to enable more accurate estimation of direct medical cost

- Participants' experience, which includes participant satisfaction during care delivery, knowledge about ANC, and attendance, are used to access the fidelity of the participants to the intervention and their uptake of the intervention content

4.2. Data collection methods

Data collection is planned as per Appendix 1. Before randomisation, we will collect the sociodemographic characteristics of the participants through a survey.

Birth outcomes, such as SCN/NICU stays, will be extracted from the medical records of the participants after 12-week post-childbirth and one-year postpartum. The Australian health system uses a national electronic health records system called My Health Records (MHR), which covers over 90% of Australians (61,62) and has been used in several RCTs for data collection (63–65). Thus, it is easier to acquire clinical outcome data and cost estimates by accessing the medical records in MHR. For those without MHR, we will collaborate with Queensland Health makes it easier to coordinate the spaces and personnel for the trial and to estimate the cost of the intervention and collect the outcome data.

Participant satisfaction and knowledge about ANC will be assessed via online surveys after Sessions 1, 5, 10, plus 12-weeks post childbirth and one-year postpartum, on the Expect with Me platform, which is integrated with a real-time data collection system.

The details of each session, including locations, dates, facilitators, sociodemographic characteristics of participants and session attendance statuses, will be updated by the interventionists at each site after each session on the web platform.

4.3. Measurement instruments

Sociodemographic questionnaire collected patient data of the income group, smoking status (59), age (60), marriage status, and ethnicity. These characteristics are known to affect birth outcomes, according to previous studies.

Participant satisfaction will be measured using the questionnaire developed by Anderson et al. (2013) (66). It uses the four-point Likert scale to measure the perceived reality and the subjective importance of the medical and emotional aspects of ANC, respectively. They can choose from four options ranging from 'completely disagree' to 'completely agree' for perceived reality and choose from four options ranging from 'unimportant' to 'very important' for subjective importance. They reported a Cronbach's alpha of 0.88 for perceived reality, and 0.86 for subjective importance, which shows high internal consistency. The approach has been used for assessing participant satisfaction with ANC (66,67).

A dedicated antenatal knowledge questionnaire will be adapted from the previous national survey (68) and will be tested in a pilot study for its validity.

4.4. Target Population

Research staff will refer pregnant women attending an initial prenatal care visit at a local general practice or recruit them directly. A 10-dollar voucher will be provided upon successful recruitment and completion of the programme. The inclusion criteria include:

1. Pregnancy at less than 13 weeks' gestation;
2. Pregnancy is not considered high-risk;
3. Capability to read and communicate in English;
4. Willingness to participate in group prenatal care.

4.5. Monitoring and Reporting

The research team will monitor the participant attendance, as recorded on the Expect with Me platform, to ensure the trial follows the theory of change. The participants may also track their own attendance and their responses on the platform and provide feedback to the research team. Interventionists will also complete a fidelity checklist after each group session to assess the delivery of the intervention and ensure fidelity. In case of unexpected events, the team will inform the ethical committee and relevant stakeholders to address the issue promptly.

4.6. Data analysis

An intention-to-treat analysis will be used to evaluate the intervention via Stata at the end of the study. Binary outcomes, such as SCN/NICU admissions, will be aggregated to calculate their relative risks. Chi-square tests or Fisher's exact tests will be performed to whether the differences are statistically significant. Regression models will be used to compare continuous outcomes between the two arms, adjusting for potential confounders and clustering effects. Descriptive statistics will also be presented for a comprehensive overview.

5. Ethical Considerations

The trial will be conducted following the National Statement on Ethical Conduct in Human Research (69) and the Guidelines for Ethical Conduct in Aboriginal and Torres Strait Islander Health Research (70). These documents provide comprehensive guidance on ethical considerations in research involving human participants, with particular attention to the unique cultural context of Aboriginal and Torres Strait Islander communities. Before commencing the trial, ethical approval will be obtained from the National Research and Evaluation Ethics Committee, the designated human research ethics committee (HREC) for clinical studies conducted in general practitioner and primary care settings in Australia. Since the study may involve Aboriginal participants, ethical approval will also be sought from the HREC of the Australian Institute of Aboriginal and Torres Strait Islander Studies. The trial will be registered with the Australian New Zealand Clinical Trials Registry (71) before the enrolment of the first participant. This registration will ensure that the study design, methodology, and outcomes are publicly available for scrutiny. The results will be published per CONSORT 2010 (72).

5.1. Informed Consent

All participants will be provided with detailed information about the study, including its purpose, procedures, potential risks and benefits to both mothers and neonates (73), and their rights as participants. They will be given sufficient time to review this information and ask any questions before deciding to participate. Written informed consent will be obtained from each participant, and they will be free to withdraw from the study at any time without any negative consequences. Consent forms will follow the above-mentioned national guidelines and guidance from HRECs.

Informed consent will be obtained separately to access the participant's health records kept by Queensland Health or in MHR, for data extraction from medical records. This will fulfil the requirements of the Australian Digital Health Agency, the government agency responsible for MHR, and Queensland Health, the local health authorities.

5.2. Confidentiality and Anonymity

To ensure the confidentiality and anonymity of the study participants, the collected data will be presented in aggregate form in publications and for future research purposes. This implies that there will be no personal identifiers included. As the trial uses a web platform made secure under the Health Insurance Portability and Accountability Act of 1996 in the United States, the web platform will be adapted to The Privacy Act 1988 in Australia to comply with local law.

A previous trial shows that information sharing during group sessions may raise privacy concerns (26). Thus, the participants are advised not to discuss the topics discussed during the group session or share them outside of the group (74). Meeting records in the Expect with Me system will only allow the participants to access their own data, e.g. attendance, and will not reveal the identity of other participants to ensure confidentiality.

To avoid unauthorised access to personal information, participant data will be de-identified and the research team will maintain electronic data in a secure, password-protected database that is only accessible to team members. Physical data will be stored in locked drawers at the research facility. All data that could potentially identify the participants will be destroyed ten years after the study is completed.

5.3. Minimising Risk

Based on current evidence, the usual care provided by the Australian health system are one of the safest in the world, while Expect with Me has shown to be safer than usual care in a previous cohort study in the United States (51). Thus, the trial exposes none to higher-than-normal risk. As the decision for SCN/NICU admission is made independent of the reimbursement policy, but based on the medical need (57), choosing SCN/NICU admission as the primary outcome will not encourage or discourage SCN/NICU admissions.

As the intervention involves online components, we anticipate there be digital divide for certain participants, which could be a risk for the trial. Thus, we will inform the participants of the free Wi-Fi provided by Queensland Health or other public providers and how to use it. Participants will also receive any necessary technical support to ensure that they are able to access and participate in the intervention. Additionally, we will conduct a needs assessment at the beginning of the trial to identify any potential barriers to participation in the intervention and work with participants to address these barriers.

We acknowledge that despite our efforts to minimise the impact of the digital divide, it may not be possible to completely eliminate this issue. We will also consider the potential limitations of the study in terms of generalisability to populations who may not have equal access to the necessary technology or technical support.

5.4. Justice

Randomised evaluation is necessary due to the uncertainty surrounding the intervention's effects in Australia. If the intervention is proven effective and cost-effective, the evidence will provide a solid foundation for scaling it nationwide.

A lottery design is used to ensure fairness. Participants will be told that both group and traditional ANC are considered safe in Australia. However, group ANC was tested in the United States for clinical outcomes, but not in the Australian public health system. Participants are informed of the risk when Expect with Me is not as effective as expected. Participants will be told that the cost will be reimbursed no matter what group they are assigned to.

The study will not involve deception. Participants will be told the pros and cons of usual care and group ANC. Control-group participants will have access to the intervention content on the Expect with Me platform after the trial.

6. Acknowledgement

We thank the Queensland Maternity and Neonatal Clinical Network for providing the data used in this protocol.

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8. Appendices

Appendix 1. Timetable for Expect with Me

Session	Themes and topics	
Recruitment and baseline measurement (before 13 weeks): • Demographic questionnaire • Antenatal Risk Questionnaire		
1 (13–17 weeks)	**You're a healthy mom** • Healthy lifestyle and eating habits during pregnancy • Weight management • Routine prenatal testing and emergencies • Knowledge about blood pressure and weight numbers	
2 (17–21 weeks)	**Staying healthy and strong through change** • Knowledge about the development of a baby • Oral health during pregnancy • Emotion, movement, sleep management • Safety at home, work and play	
3 (21–24 weeks)	**Breastfeeding = Healthy Babies and Healthy Moms** • Benefits, barriers and basics of breastfeeding • How to choose a paediatric provider • How to seek support	
4 (25–29 weeks)	**Healthy moms building healthy relationships** • Knowledge about Gestational Diabetes Testing • Relationship management during pregnancy • Sexual health and STD prevention during pregnancy • Timing for pregnancy	
5 (27–31 weeks)	**Healthy moms and healthy labour** • Signs, stages of labour • Why and how to do foetal heart rate monitoring • What to do during labour and Caesarean birth	
Follow-up 1 (between 27-33 weeks): • Participant satisfaction questionnaire • Participant knowledge questionnaire		
6 (29–33 weeks)	**Healthy labour** • Stages of labour and what follows • Available labour and delivery decisions • Available labour and delivery policies and options • Sexual health and STD prevention during pregnancy	
7 (31–35 weeks)	**Healthy labour and healthy relationships** • What to do when in hospital stay and returning home • Negotiate in relationship management • Knowledge about Group B Strep testing and prevention	
8 (33–37 weeks)	**Taking care of mom and baby** • How to take care of your baby and yourself • How to choose a paediatric provider • Goalsetting for relationship management	

Session	Themes and topics	
9 (35–39 weeks)	**Preparing for a Healthy Future** • Breastfeeding ABC • Emotion management after pregnancy • Safety management after pregnancy	
10 (37+ weeks)	**Build a healthy future** • Available choice for day-care or alternative care • What to do before going back to work • How you can get support • Choose when to get pregnant again • Goalsetting for relationship management	
Follow-up 2 (For healthy outcomes: 12-weeks post childbirth; For adverse outcomes: one-year postpartum): • Participant satisfaction questionnaire • Participant knowledge questionnaire • Birth outcomes, extracted from medical records • Medical resource utilisation, extracted from medical records		

Note: Adapted from Cunningham et al. (2017)(31).

Modification 1: Recruitment deadline changed from 24 weeks of gestation to 13 weeks of gestation to ensure the participants can attend all sessions.

Modification 2: We follow the previously described trial design for Expect with Me for data collection (31), except for provider surveys and focus group discussions, as we focus on clinical outcomes and the cost-effectiveness of the intervention.

Appendix 2. Sample size calculation

```
. power twoproportions 0.202 0.131, k1(20(1)50) kratio(1) rho(0.05) cluster

Performing iteration ...

Estimated cluster sizes for a two-sample proportions test
Cluster randomized design, Pearson's chi-squared test
H0: p2 = p1  versus  Ha: p2 != p1
```

alpha	power	K1	K2	M1	M2	delta	p1	p2	rho
.05	.8	20	20	.	.	-.071	.202	.131	.05
.05	.8	21	21	.	.	-.071	.202	.131	.05
.05	.8	22	22	908	908	-.071	.202	.131	.05
.05	.8	23	23	283	283	-.071	.202	.131	.05
.05	.8	24	24	168	168	-.071	.202	.131	.05
.05	.8	25	25	119	119	-.071	.202	.131	.05
.05	.8	26	26	92	92	-.071	.202	.131	.05
.05	.8	27	27	76	76	-.071	.202	.131	.05
.05	.8	28	28	64	64	-.071	.202	.131	.05
.05	.8	29	29	55	55	-.071	.202	.131	.05
.05	.8	30	30	49	49	-.071	.202	.131	.05
.05	.8	31	31	44	44	-.071	.202	.131	.05
.05	.8	32	32	40	40	-.071	.202	.131	.05
.05	.8	33	33	36	36	-.071	.202	.131	.05
.05	.8	34	34	33	33	-.071	.202	.131	.05
.05	.8	35	35	31	31	-.071	.202	.131	.05
.05	.8	36	36	29	29	-.071	.202	.131	.05
.05	.8	37	37	27	27	-.071	.202	.131	.05
.05	.8	38	38	25	25	-.071	.202	.131	.05
.05	.8	39	39	24	24	-.071	.202	.131	.05
.05	.8	40	40	23	23	-.071	.202	.131	.05
.05	.8	41	41	22	22	-.071	.202	.131	.05
.05	.8	42	42	21	21	-.071	.202	.131	.05
.05	.8	43	43	20	20	-.071	.202	.131	.05
.05	.8	44	44	19	19	-.071	.202	.131	.05
.05	.8	45	45	18	18	-.071	.202	.131	.05
.05	.8	46	46	17	17	-.071	.202	.131	.05
.05	.8	47	47	17	17	-.071	.202	.131	.05
.05	.8	48	48	16	16	-.071	.202	.131	.05
.05	.8	49	49	15	15	-.071	.202	.131	.05
.05	.8	50	50	15	15	-.071	.202	.131	.05

```
. power twoproportions 0.202 0.131, m1(23) mratio(1) rho(0.001 0.01 0.05) cluster

Performing iteration ...

Estimated numbers of clusters for a two-sample proportions test
Cluster randomized design, Pearson's chi-squared test
H0: p2 = p1  versus  Ha: p2 != p1
```

alpha	power	K1	K2	M1	M2	delta	p1	p2	rho
.05	.8	20	20	23	23	-.071	.202	.131	.001
.05	.8	23	23	23	23	-.071	.202	.131	.01
.05	.8	40	40	23	23	-.071	.202	.131	.05

Appendix 3. Estimation of available facilities

Existing sites: 108

By consulting Queensland Health, we learn that there are 108 existing sites providing ANC under Queensland Health

Community centre: 143+

Another options for intervention delivery is the community centres, which has been used to deliver various interventions in other studies (75–77). In Queensland, there are 143+ of such facilities, mostly available for room hire. Thus, this ensures we can have at least 200 sites available.

General practices: unclear

We may also work with a local GP to provide the care. In 2019, there are 37,000 general practitioners (GPs) working in 8,147 general practices across Australia (78). As around 22% GPs are in Queensland,(79) we estimated there be 1,700 general practices in Queensland. We estimate that there are at least 1700 general practices in Queensland. Yet, how many GPs are able to provide the space for the intervention is unclear and require further survey.

Current need for facilities: less than 80

However, based on the existing information, we will have enough sites for the interventions, as we may still estimate the sample size (see Appendix 2 and Figure 3).